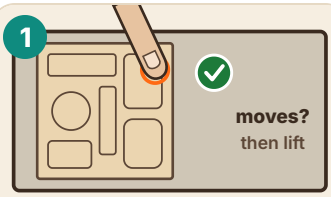
# SOP 3 — Unload and pop-out check

**When:** Every job, before the sheets leave the laser **Time:** 2 to 4 min per job (Estimate)
**Who:** Laser operator

**You need:** Bridge map for the sheet ( `kits/<kit>/popout/` ) · A lamp or window · Planner tablet




### Before you lift: test a corner part.

Press lightly on a small part near each corner. It should move a little.

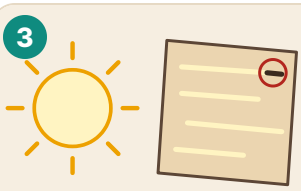
- Moves: go on. Stuck: next step.




### Stuck? Run the cut layer again.

Do not move the sheet. Turn off engrave and score. Run cut once.

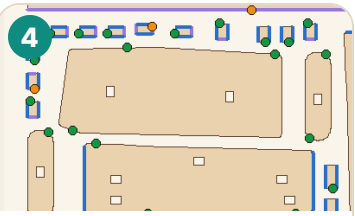
- Still stuck: weak power or dirty lens (SOP-5).




### Lift the sheet flat. Hold it to the light.

Use both hands and do not bend it. Light should show along every cut line.

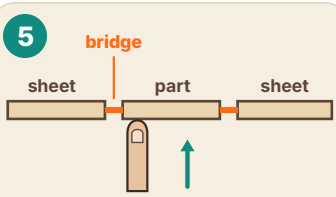
- A dark spot in a line is an uncut spot. Log "uncut".




### Compare with the bridge map.

Green dots show the bridges. Each part hangs by 1 to 4 bridges, 0.5 mm wide.

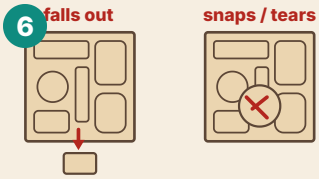
- Blue edges slide into other parts: never a bridge there.




### Pop test: first sheet of each batch.

Press from the back, right next to a bridge. One thumb should pop the part.

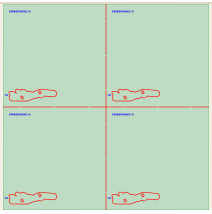
- Put the test parts in the kit's small-parts bag.




### Falls out alone, or tears when pushed?

Falls out: the kerf is too wide, lower cut power 5%. Tears or needs force: clean the lens, check focus.

- Recut the coupon bridges (SOP-5). Log "bridge".




### Half and quarter sheets: snap the tiles.

Bend gently along the long separation cut. Each tile goes to a different kit.

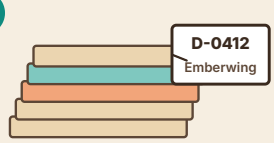
- Clean edge, no part broken off at the line.




### Check the show faces.

A light smoke stain sands off. A dark scorch on a visible part means remake the sheet.

- Log "burn" if you remake it.




### Stack flat by kit, with a label.

Label: kit, batch, date. Then mark the batch Cut in the planner.

- Sheet letters match the job name.

## Stop the batch and fix first if

- Two or more uncut spots on one sheet, or parts falling out on two sheets in a row.
- A crack runs through a part (bad sheet: set the stack of that lot aside).

Hartwood Designs, SOP-3, Rev A, 2026-10-12. Times and settings marked Estimate are planning numbers until measured on the real machine. Machine details follow the OMTech manual when it differs from this sheet.